In[79]:=
```
Remove["Global`*"]
```
去除
```
Γself[a_, b_, c_] := Γ^a_.._b.c (*Γself表面Γ这个符号*)
zuobiao = {x, y} (*待输入*);
g0 = {(2 / (1 - (x^2 + y^2)))^2, (2 / (1 - (x^2 + y^2)))^2};
g = DiagonalMatrix[g0];(*待输入*)
```
对角矩阵
```
dimg = Dimensions[g][[1]];
```
维数
```
ing = Inverse[g];
```
逆

```
(*Γ的计算*)
cΓself[a_, b_, c_] := Block[{μ = Position[zuobiao, a][[1, 1]],
```
块 位置
```
    ν = Position[zuobiao, b][[1, 1]], σ = Position[zuobiao, c][[1, 1]], k},
```
位置 位置
```
  Simplify[Sum[(1 / 2) * ing[[μ, k]] *
```
化简 求和
```
      (D[g[[ν, k]], c] + D[g[[σ, k]], b] - D[g[[σ, ν]], zuobiao[[k]]]), {k, dimg}]]]
```
偏导 偏导 偏导
```
Table[Γself[μ, ν, σ] == cΓself[μ, ν, σ],
```
表格
```
  {μ, zuobiao}, {ν, zuobiao}, {σ, zuobiao}] // TableForm(*列表*)
```
表格形式

```
(*给出测地线方程，带参数λ*)
cedixian =
  Table[zuobiao[[k]]'' == -Sum[cΓself[zuobiao[[k]], zuobiao[[i]], zuobiao[[j]]] * zuobiao[[i]]' *
```
表格 求和
```
      zuobiao[[j]]', {i, dimg}, {j, dimg}], {k, dimg}];

cedixian = cedixian /. Join[Table[zuobiao[[k]] → zuobiao[[k]][λ], {k, dimg}],
```
连接 表格
```
    Table[zuobiao[[k]]' → zuobiao[[k]]'[λ], {k, dimg}],
```
表格
```
    Table[zuobiao[[k]]'' → zuobiao[[k]]''[λ], {k, dimg}]]
```
表格

```
(*给出指数映射*)
exp[q_, u_] :=
 Block[{sol}, sol = NDSolve[Join[cedixian, Table[zuobiao[[k]][0] == q[[k]], {k, dimg}],
```
块 数值… 连接 表格
```
     Table[zuobiao[[k]]'[0] == u[[k]], {k, dimg}]], Table[zuobiao[[k]], {k, dimg}], {λ, 0, 1}];
```
表格 表格
```
  Table[zuobiao[[k]][1], {k, dimg}] /. sol[[1]]
```
表格
```
 ]
```

```
exp[{-0.5, -0.5}, {0.5, 0.3}]
exp[{0.2, -0.4}, {-0.2, 0.8}]
```

Out[87]//TableForm=

$$\Gamma^{x}{}_{x.x} == -\frac{2x}{-1+x^2+y^2} \qquad \Gamma^{x}{}_{y.x} == -\frac{2y}{-1+x^2+y^2}$$
$$\Gamma^{x}{}_{x.y} == -\frac{2y}{-1+x^2+y^2} \qquad \Gamma^{x}{}_{y.y} == \frac{2x}{-1+x^2+y^2}$$
$$\Gamma^{y}{}_{x.x} == \frac{2y}{-1+x^2+y^2} \qquad \Gamma^{y}{}_{y.x} == -\frac{2x}{-1+x^2+y^2}$$
$$\Gamma^{y}{}_{x.y} == -\frac{2x}{-1+x^2+y^2} \qquad \Gamma^{y}{}_{y.y} == -\frac{2y}{-1+x^2+y^2}$$

Out[89]=

$$\left\{x''[\lambda] == \frac{2x[\lambda]\,x'[\lambda]^2}{-1+x[\lambda]^2+y[\lambda]^2} + \frac{4y[\lambda]\,x'[\lambda]\,y'[\lambda]}{-1+x[\lambda]^2+y[\lambda]^2} - \frac{2x[\lambda]\,y'[\lambda]^2}{-1+x[\lambda]^2+y[\lambda]^2},\right.$$
$$\left. y''[\lambda] == -\frac{2y[\lambda]\,x'[\lambda]^2}{-1+x[\lambda]^2+y[\lambda]^2} + \frac{4x[\lambda]\,x'[\lambda]\,y'[\lambda]}{-1+x[\lambda]^2+y[\lambda]^2} + \frac{2y[\lambda]\,y'[\lambda]^2}{-1+x[\lambda]^2+y[\lambda]^2}\right\}$$

Out[91]=

```
{0.376057, -0.297723}
```

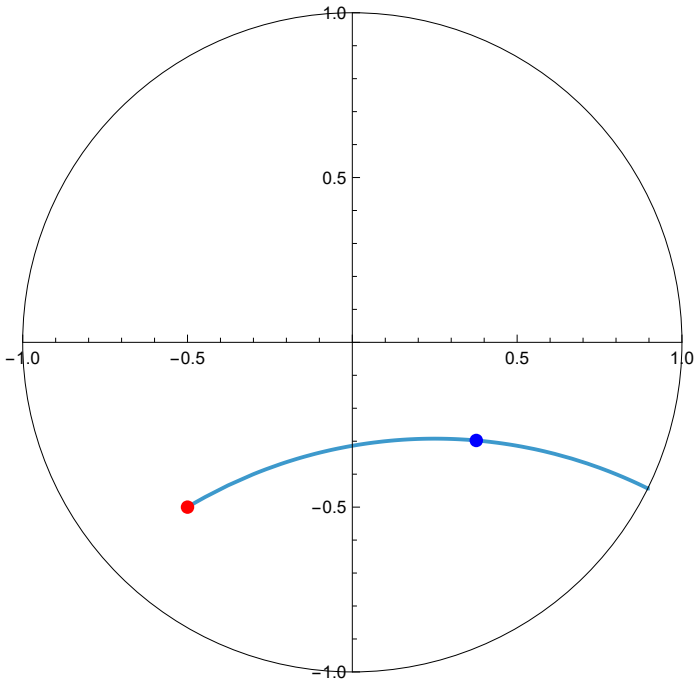
Out[92]=

```
{0.0776407, 0.521556}
```

In[76]:=

```
(*测地线绘图展示eg: (庞加莱圆盘) *)
drawGeodesic[q_, u_] := Module[{sol},
                        模块
  sol = NDSolve[Join[cedixian, Table[zuobiao[[k]][0] == q[[k]], {k, dimg}], Table[
        数值…  连接              表格                                          表格
      zuobiao[[k]]'[0] == u[[k]], {k, dimg}]], Table[zuobiao[[k]], {k, dimg}], {λ, 0, 5}];
                                                  表格
  ParametricPlot[Evaluate[Table[zuobiao[[k]][λ], {k, dimg}] /. sol[[1]]],
  绘制参数图          计算        表格
    {λ, 0, 5}, PlotRange → {{-1, 1}, {-1, 1}},
                绘制范围
    Epilog → {Circle[], PointSize[Large], Red, Point[q], Blue, Point[exp[q, u]]}]]
  绘制主…  圆          点的大小  大        红色  点          蓝色  点
drawGeodesic[{-0.5, -0.5}, {0.5, 0.3}]
drawGeodesic[{-0.5, -0.5}, {1, 0.6}]
```

Out[77]=



Out[78]=

